

EFB = empty fruit bunch.

Source: Murdiyatmo (2021).

However, the main obstacle with producing second-generation bioethanol is the cost of enzymes. Murdiyatmo (2021) stated that, at the pilot scale, the cost of enzymes is very high, i.e. Rp18,000 per litre of ethanol produced. Some studies provided the cost of enzymes in the US. NREL (2011), for instance, estimated that the cost of enzymes to produce second-generation bioethanol in the US was equivalent to around \$0.34 per gallon or Rp1,529² per litre of ethanol produced, i.e. less than one-tenth of the cost of enzymes in Indonesia.

In the next sub-sections, we analyse biodiesel and bioethanol introduction in Indonesia. In each sub-section, we first discuss the current supply and demand of the biofuels and the related conventional transport fuel. Second, we estimate the conventional transport fuel, i.e. gasoline and diesel fuel demand in road transportation during the period of 2020–50. Third, we estimate the volume of pure biofuel (fatty acid methyl ester [FAME]/biodiesel and bioethanol) needs in scenarios, and in the amount of feedstock, i.e. CPO in biodiesel and molasses in bioethanol needed to meet the demand required in each scenario.

2.1. Diesel and biodiesel use

The consumption of diesel fuel in Indonesia, used primarily for road freight transport, fluctuated between 2010 and 2019 as it correlated with the economic condition (Table 2.8). Diesel consumption in the industry sector decreased significantly, around 10% per year between 2010 and 2019, resulting from the shift to another energy type. During the same period, with some fluctuations, diesel production increased at 3.6% annual growth rate, while imports were cut by half from nearly 13 billion litres in 2010 to nearly 6.5 billion litres in 2018. The biodiesel blending rate increased from only 1% in 2010 to nearly 20% in 2019, representing a growing level of mandatory biodiesel programmes. Apparently, diesel imports dropped with the increase of the biodiesel (B100) blending rate.

² Assuming average inflation rate of 2% between 2011 and 2021 and an exchange rate of \$1 = Rp14,131.